

Graphplay

A machine-checked theory of equitable-partition quantum walks

Verified attention-complexity reduction, an $ML \rightarrow$ chip compiler, and falsifiable hardware experiments

Graphplay Crew — pitch draft, May 2026

Abstract

We give a Lean 4 + Mathlib formalization of equitable-partition quantum walks and use it to prove five things that, to our knowledge, have not been proven (let alone machine-checked) before. (1) Structured attention compiles to a continuous-time quantum walk (CTQW) Hamiltonian whose equitable quotient evaluates the attention map **exactly** — not approximately — in $O(n \cdot r)$ work where the naive cost is $O(n^2)$; the correctness identity `blockAttentionApply_eq_fullAttentionApply` is sorry-free. (2) A typed $ML \rightarrow$ chip compiler whose semantics-preservation theorem `compile_denote_commutes` is proven: the compiled quotient computes the same function the source program denotes, and is strictly cheaper. (3) Verified quantum-search advantage on a **sparse, physically buildable** host: the Boolean hypercube’s Hamming-distance equitable quotient collapses marked search to a $(d\{+\}1)$ -dimensional chain, with sparsity and the exact equitable reduction proven axiom-clean (`hypercube_sparse_search_advantage`); we are honest that the $O(\sqrt{N})$ **timing** clause, and the $d > 4$ lattice threshold (`lattice_search_dimension_threshold`), are deferred to the Chakraborty–Novo–Roland spectral criterion. (4) Verified spectral disassembly of real chips — IBM heavy-hex (`dataFlagQuotient_eigenvalues`) and the Microsoft Majorana-1 parity sectors (`sectorProjector_sum`) — as equitable quotients. (5) A falsifiable on-device experiment (`compiled_experiment_prediction`) emitted by the same compiler, predicting a key population from a verified amplitude on IBM Heron hardware.

The honest claim. The **mathematics** of equitable-quotient quantum walks is not ours: it is Godsil, Bachman–Tamon, Ide–Narimatsu, Childs–Goldstone, and Janmark–Meyer–Wong (§ 4). The walk-as-attention idea is concurrent classical work (GQWformer, CTQWformer). Our contribution is the **certificate**: the first machine-checked, end-to-end stack from an exact verified attention reduction, through a typed hardware compiler with a proven commuting semantics, to a falsifiable on-chip prediction. We seek collaborators.

1. Five results, stated precisely

Everything below is a named Lean theorem. We mark each as *proven* (sorry-free), *proven core* / *deferred clause* (the load-bearing identity is sorry-free, an explicitly flagged dynamical/spectral clause is deferred), or *open*.

*Result 1 — Exact $O(n^2) \rightarrow O(n \cdot r)$ attention. **proven**.* Cast a (symmetrized) attention pattern as a weighted token graph and suppose it is block-constant under an equitable partition into r cells. Then the attention apply factors through the $r \times r$ symmetric quotient and computes the **same output vector** in $O(n \cdot r)$ instead of $O(n^2)$ work:

$$\text{blockAttentionApply}(A, B, \text{cell}, V) = \text{fullAttentionApply}(A, V).$$

This is `blockAttentionApply_eq_fullAttentionApply` in `Graphplay/Integrations/AttentionComplexity.lean` — sorry-free, with the forward apply, the backward pass (`blockGrad_apply`), and a training step (`training_step_linear_under_equitable`) all linear in n . The reduction is **exact**: no rank truncation, no kernel approximation. (The separate `attention_quantum_composition` carries an honest sorry on a deep quantum-rate clause; the structural reduction does not depend on it.)

*Result 2 — A verified $ML \rightarrow$ chip compiler. **proven**.* `Graphplay/Integrations/TransformerDSL.lean` defines a small typed language of transformer programs with a denotational semantics `denote` and a compiler `compile` into the quotient representation. The keystone is sorry-free:

$$\text{denote}(P) = \text{denote}_{\text{quotient}}(\text{compile}(P)) \quad (\text{compile_denote_commutes}).$$

The functor `compiler_guarantee` packages this with the complexity drop (`compiled_apply_le_naive`): the compiled program is **semantics-preserving and strictly cheaper**. This is the “functor stack” — a compiler whose correctness is a theorem, not a test suite. (catgrad, the categorical auto-differentiation framework from hellas.ai, is the intended practical frontend for emitting these programs from real models.)

*Result 3 — Search advantage on a buildable sparse host. **proven core / deferred timing**.* `hypercube_sparse_search_advantage` (`Applications/SparseSearch.lean`) proves, axiom-clean, that the Boolean hypercube Q_d is (a) d -regular with $d = \log_2 N - \log\text{-degree}$, hence sparse and physically realizable, unlike K_N ’s degree $N\{-\}1$ — and (b) that its Hamming-distance partition is equitable and collapses marked CTQW search exactly to a $(d\{+\}1)$ -dimensional chain. The remaining $O(\sqrt{N})$ **timing** clause is the single honest sorry, deferred to the Chakraborty–Novo–Roland spectral-ratio criterion (arXiv:2004.12686) — the same dynamical core deferred by the K_n flagship. The general lattice story is stated as `lattice_search_dimension_threshold`: a d -dimensional lattice supports the advantage iff $d > 4$ (Childs–Goldstone); the statement is written, its spectral (forward) half deferred. The exact finite- n K_n amplitude `quantum_search_exact_amplitude` is fully proven.

*Result 4 — Verified disassembly of real chips. **proven**.* We model two real devices as equitable quotients and check the spectral data in Lean. The IBM heavy-hex lattice has a 2-cell data/flag equitable partition; `dataFlagQuotient_eigenvalues` (`Applications/IBMHeavyHex.lean`) computes the quotient spectrum, from which cell-uniform PST timing on the chip’s **native** couplings is read off. Microsoft’s Majorana-1 tetron has a joint-parity-sector partition; `sectorProjector_sum` (`Applications/MajoranaOne.lean`) proves the sector projectors resolve the identity, i.e. the parity sectors are a genuine orthogonal decomposition. (Majorana-1 honesty: present hardware **simulates** Majorana modes rather than realizing native topological qubits; this file is a modeling exercise, flagged as such.)

*Result 5 — A falsifiable on-Heron experiment. **proven structural form**.* From the heavy-hex disassembly the compiler emits a concrete prediction: `compiled_experiment_prediction` (`Applications/CompileML.lean`) ties a predicted measured key-population to the squared amplitude of the verified quotient walk (`predictedKeyPopulation_eq_amplitude_sq`). A companion witness `compiled_positive_breakingScore_exists` exhibits a configuration whose breaking-score is nonzero — i.e. a **falsifier**: an experiment that, run on IBM Heron, would disconfirm the prediction if the cell-uniform structure is broken. A verified prediction with a built-in way to be wrong is the experimental teeth of the whole stack.

2. The spine: equitable-quotient lifts, mechanized

All five results ride on one structural skeleton, formalized once and reused. Given an operator A (an attention pattern, a search Hamiltonian, a chip coupling matrix) and an **equitable partition** π with characteristic isometry P and symmetric quotient A/π :

1. *Spectral lift.* $\text{spec}(A/\pi) \subseteq \text{spec}(A)$, with cell-constant eigenvectors lifting along P .
2. *Dynamical restriction.* $\exp(-itA)$ commutes with the cell-uniform projector and acts on its range as $\exp(-it(A/\pi))$.
3. *Primitive transport.* For cell-uniform initial/target data, PST / mixing / search holds on (A, π) iff it holds on the small matrix A/π — this is `cellUniformPST_iff_quotientPST` (`PST/QuotientIff.lean`), our mechanization of the Bachman–Tamon quotient-iff theorem.

The point of the formalization is that **the same three lemmas** are reused verbatim by the attention reduction (Result 1), the search collapse (Result 3), and the chip disassemblies (Results 4–5). The library builds end-to-end on a current Mathlib; the load-bearing theorems named in § 1 are sorry-free or have explicitly-flagged single deferred clauses. Larger “dowsing-rod” extension files (graphon limits, ∞ -categorical towers) carry precise statements with proofs in progress — these are the research frontier, not the load-bearing claims.

3. Why a certificate, in this domain, is the whole point

Two failure modes plague quantum-walk-for-ML and quantum-hardware claims: silently approximate “exact” reductions, and speedups that evaporate under state-preparation/readout caveats (Aaronson, **Read the fine print**). A machine-checked proof closes the first: `blockAttentionApply_eq_fullAttentionApply` **cannot**

be an approximation that was rounded into an equality, because Lean would reject it. And by tracking every sorry, we make the second mode auditable: the deferred clauses (CNO timing, the $d > 4$ spectral half) are named and isolated, so a reader sees exactly where the physics input is assumed rather than proven. The honesty is itself the contribution: a transparent proven/deferred ledger is something the empirical and pencil-and-paper literature structurally cannot provide.

4. Honest novelty --- what is prior art, what is ours

We are a **certificate**, not first contact. The mathematics is established and we attribute it explicitly:

Idea	Source (prior art — we formalize, not originate)
Quotient-iff PST under equitable partition	Bachman–Tamon et al. 2012 (arXiv:1108.0339); Godsil survey 2011 (arXiv:1102.4898)
CTQW search collapses to a small quotient via equitable partition + PST	Ide–Narimatsu 2022 (arXiv:2209.07688) — closest prior work to our spine
$O(\sqrt{N})$ spatial search on K_n / hypercube	Childs–Goldstone 2004 (quant-ph/0306054)
Fast search needs only local (equitable), not global, symmetry	Janmark–Meyer–Wong 2014 (PRL 112 210502 / arXiv:1403.2228)
$d > 4$ lattice search threshold	Childs–Goldstone; CNO spectral criterion 2020 (arXiv:2004.12686)
Quantum walk as transformer attention bias (classical)	GQWformer 2024 (arXiv:2412.02285), CTQWformer 2026 (arXiv:2605.09486) — <i>concurrent</i>

GQWformer and CTQWformer are concurrent **classical** attention-as-walk work: they compute a walk classically as an inductive bias for graph classification, with no exactness guarantee, no complexity reduction, no hardware, and no proof. Our wedge is orthogonal to theirs along three axes simultaneously:

exactness (the reduction is an identity, not a heuristic feature) $\otimes$ **verification** (every claim is a Lean theorem, with a transparent sorry-ledger) $\otimes$ **hardware** (a typed compiler emitting a falsifiable experiment for a named chip).

No prior work claims, let alone machine-checks, the exact $O(n^2) \rightarrow O(n \cdot r)$ equitable-attention reduction, the typed ML $\rightarrow$ chip compiler with a commuting semantics, or the compiler-emitted falsifiable PST experiment. That stack is ours and, as far as we can determine, uncontested.

5. Open problems and the research program

We are explicit about what is **not** done:

Deferred dynamical/spectral clauses (named, isolated). The $O(\sqrt{N})$ CNO timing for hypercube/lattice search; the forward (spectral) half of `lattice_search_dimension_threshold`. These reduce to the Chakraborty–Novo–Roland ratio criterion, which we have stated but not yet mechanized.

Conjectural with worked examples. Conjecture 9.3 — a consistent sequence (G_n, π_n) admits both a graphon quasi-infinite limit and a strict chiral mixing speedup iff its limit Bose–Mesner algebra coincides with the limit partition-projector algebra. Weak version expected true, strong version expected false; we have both-sides-false witnesses and one candidate counterexample.

Frontier towers (scaffold). The ∞ -categorical / sheaf-theoretic envelopes (quasi-infinite walks as filtered colimits; topological protection as a sheaf H^1 -class) have precise statements awaiting Mathlib infrastructure (operator-system, sheaf-of- C^* -algebra, and ∞ -category libraries) and the Dirac/continuum limit. These are multi-year items, marked open.

6. How to get involved

The repository builds with `lake build` on a current Mathlib. By audience:

- **CTQW / quantum-walk mathematicians.** Discharge a deferred spectral clause — the CNO timing in Applications/SparseSearch.lean is the highest-value single hole and would complete the search-advantage headline.
- **ML / systems people.** Wire catgrad (hellas.ai) as the frontend: emit TransformerPrograms from real models so the verified compiler runs on production attention patterns.
- **Quantum-hardware engineers.** Run compiled_experiment_prediction on IBM Heron; adversarially read the heavy-hex and Majorana-1 disassemblies.
- **Mathlib contributors.** The operator-system / quantum-graph and graphon- spectrum infrastructure are clean upstream candidates.

Each is self-contained, and each is a paper, a Mathlib PR, or a worked example we will fold into the spine.

Proven anchors: blockAttentionApply_eq_fullAttentionApply, compile_denote_commutates, compiler_guarantee, quantum_search_exact_amplitude, cellUniformPST_iff_quotientPST, dataFlagQuotient_eigenvalues, sectorProjector_sum, compiled_experiment_prediction (structural). *Proven core / deferred clause:* hypercube_sparse_search_advantage, lattice_search_dimension_threshold. *Open:* Conjecture 9.3, ∞ -categorical towers, CNO timing, Dirac limit. *Contact:* open an issue on the repository. Collaborators welcome at any level.